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git.

2026-04-03

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- 1032253529@rudn.ru
- https://PALopatchenko-lab.github.io/ru/

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- git.

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- git.
- SSH.

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- git.
- SSH.
- PGP.

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- git.
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- git.
- SSH.
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- git.
- GitHub.

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- git.
- SSH.
- PGP.
- git.
- GitHub.
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3

1) git gh

git GitHub CLI gh.

```
palopatchenko@palopatchenko:~$ sudo dnf install git gh
[sudo] password for palopatchenko:
Updating and loading repositories:
 Fedora 43 - x86_64 - Updates
Repositories loaded.
Package "git-2.53.0-1.fc43.x86_64" is already installed.

Package                                    Arch       Version
Installing:
 gh                                        x86_64     2.87.0-2.fc43

Transaction Summary:
 Installing:         1 package

Total size of inbound packages is 11 MiB. Need to download 11 MiB.
After this operation, 38 MiB extra will be used (install 38 MiB, remove 0 B).
Is this ok [y/N]: y
[1/1] gh-0:2.87.0-2.fc43.x86_64
--------------------------------------------------------------------------------
[1/1] Total
Running transaction
```

2) git

email, CRLF.

```
palopatchenko@palopatchenko:~$ git config --global user.name "Polina Lopatchenko"
palopatchenko@palopatchenko:~$ git config --global user.email "lopatapol228@gmail.com"
palopatchenko@palopatchenko:~$ git config --global core.quotepath false
palopatchenko@palopatchenko:~$ git config --global init.defaultBranch master
palopatchenko@palopatchenko:~$ git config --global core.autocrlf input
palopatchenko@palopatchenko:~$ git config --global core.safecrlf warn
```

2: git

3) SSH-

SSH- .

```
palopatchenko@palopatchenko:~$ ssh-keygen -t rsa -b 4096
Generating public/private rsa key pair.
Enter file in which to save the key (/home/palopatchenko/.ssh/id_rsa):
Created directory '/home/palopatchenko/.ssh'.
Enter passphrase for "/home/palopatchenko/.ssh/id_rsa" (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/palopatchenko/.ssh/id_rsa
Your public key has been saved in /home/palopatchenko/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:eqHtT9i9w3ogffPbruKGrR5YQ0gioMvSwTI3RdClXjY palopatchenko@palopatchenko
The key's randomart image is:
+---[RSA 4096]----+
|  o=+.o .        |
| o ..o o .       |
|+ = . E . .      |
|.* + o . .       |
|o.. .   S.o      |
|.       +.*ooo   |
|       o =.+*.o  |
|        o ...B... |
|         .o=*ooo+o|
+----[SHA256]-----+
```

3: SSH-

4) GPG-

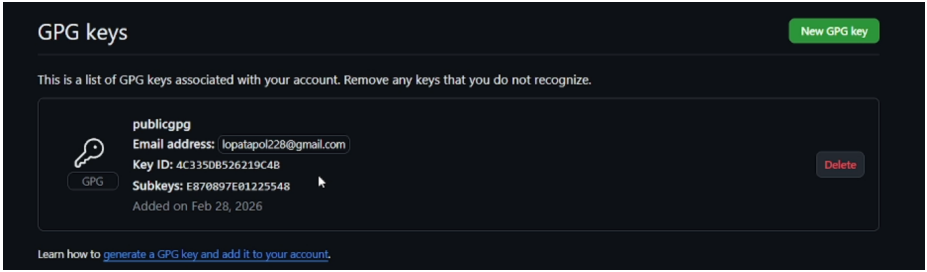
GPG- GitHub.

```
palopatchenko@palopatchenko:~$ gpg --full-generate-key
gpg (GnuPG) 2.4.8; Copyright (C) 2025 g10 Code GmbH
This is free software: you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.

gpg: directory '/home/palopatchenko/.gnupg' created
Please select what kind of key you want:
   (1) RSA and RSA
   (2) DSA and Elgamal
   (3) DSA (sign only)
   (4) RSA (sign only)
   (9) ECC (sign and encrypt) *default*
  (10) ECC (sign only)
  (14) Existing key from card
Your selection? 1
RSA keys may be between 1024 and 4096 bits long.
What keysize do you want? (3072) 4096
Requested keysize is 4096 bits
Please specify how long the key should be valid.
         0 = key does not expire
      <n>  = key expires in n days
      <n>w = key expires in n weeks
      <n>m = key expires in n months
      <n>y = key expires in n years
Key is valid for? (0) 0
Key does not expire at all
Is this correct? (y/N) y

GnuPG needs to construct a user ID to identify your key.

Real name: Polina Lopatchenko
Email address: lopatapol228@gmail.com
```

4: GPG-

5) GPG- GitHub

GitHub.



5: GPG GitHub

6)

git .

```
palopatchenko@palopatchenko:~$ git config --global user.signingkey 4C335DB526219C4B
palopatchenko@palopatchenko:~$ git config --global commit.gpgsign true
palopatchenko@palopatchenko:~$ git config --global gpg.program $(which gpg)
palopatchenko@palopatchenko:~$ git config --global --list | grep -E "user.signingkey|commit.gpgsign|gpg.program"
user.signingkey=4C335DB526219C4B
commit.gpgsign=true
gpg.program=/usr/bin/gpg
```

6:

7) gh

GitHub gh auth login.

```
palopatchenko@palopatchenko:~$ gh auth login
? Where do you use GitHub? GitHub.com
? What is your preferred protocol for Git operations on this ho
st? SSH
? Upload your SSH public key to your GitHub account? /home/palo
patchenko/.ssh/id_ed25519.pub
? Title for your SSH key: Title
? How would you like to authenticate GitHub CLI? Login with a w
eb browser
```

7: gh

8)

, .

```
palopatchenko@palopatchenko:~$ mkdir -p ~/work/study/2025-2026/"os-intro"
palopatchenko@palopatchenko:~$ cd ~/work/study/2025-2026/"os-intro"
palopatchenko@palopatchenko:~/work/study/2025-2026/os-intro$ gh repo create study_2025-2026_os-intro --te
se-directoy-student-template --public
✓ Created repository PALopatchenko-lab/study_2025-2026_os-intro on github.com
✓ https://github.com/PALopatchenko-lab/study_2025-2026_os-intro
palopatchenko@palopatchenko:~/work/study/2025-2026/os-intro$ git clone --recursive git@github.com:<owner>
tro.git os-intro
bash: owner: No such file or directory
```

8:

```
palopatchenko@palopatchenko:~/work/study/2025-2026/os-intro$ ls
os-intro
palopatchenko@palopatchenko:~/work/study/2025-2026/os-intro$ cd os-intro/
palopatchenko@palopatchenko:~/work/study/2025-2026/os-intro/os-intro$ rm package.
palopatchenko@palopatchenko:~/work/study/2025-2026/os-intro/os-intro$ echo os-int
palopatchenko@palopatchenko:~/work/study/2025-2026/os-intro/os-intro$ make
Usage:
  make <target>

Targets:
  list                          List of courses
  prepare                       Generate directories structure
  submodule                     Update submules

palopatchenko@palopatchenko:~/work/study/2025-2026/os-intro/os-intro$ git add .
palopatchenko@palopatchenko:~/work/study/2025-2026/os-intro/os-intro$ git commit
[master b18c4b9] feat(main): make course structure
 2 files changed, 1 insertion(+), 14 deletions(-)
```

9:

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- git .

- git .
- SSH GPG.

- git .
- SSH GPG.
- GitHub gh, .

- `git` .
- SSH GPG.
- GitHub `gh`, .
- .